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High modulation efficiency micro-ring modulator with low driving voltage for high speed optical interconnection

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Abstract: A high modulation efficiency depletion mode silicon micro-ring modulator with NRZ modulation rate up to 128 Gb/s has been demonstrated. The MRM utilizes a meandering doping structure to enhance the overlap factor between the carrier variation region and light field to improve modulation efficiency at 33.5 pm/V, ~38% improvement from a conventional doping design. It has a quality factor of ~5200, a high electro-optical bandwidth above 67 GHz, and can reach 110 GHz under the optical peaking effect. The drive voltage of the MRM loading a 128 Gb/s signal is 1.36 V; the corresponding power consumption is ~8.54 fJ/bit.

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1. Introduction

In the face of exponential growth in data center traffic for cloud computing, artificial intelligence (AI), and deep learning, higher bandwidth densities are urgently needed for data transfer across interconnects and fabrics. Long-serving as the foundation of data center communication, traditional copper-based interconnects now has exposed its limitations. The wire resistance leads to significant power dissipation, especially during long-distance signal transmission in large data center, causing high energy costs and thermal management difficulties. Meanwhile, the bandwidth capacity is nearing its physical limit, unable to meet the demands of current data-intensive application. Optical interconnection emerges as a promising solution. Using photons as carriers, it has advantages like low latency, minimal signal loss, high bandwidth and immunity to interference, fitting modern data center needs [1–5]. The silicon photonics (SiPh) platform is prominent, with compatibility to complimentary metal-oxide-semiconductor (CMOS) enabling cost-effective manufacturing for future upgrades [6–9]. Crucially, the electro-optical modulator at the core of optical systems converts electrical to optical signals, playing a pivotal role. Its performance directly affects the overall quality of optical interconnection, making it a focus of research and experimental development efforts to meet next-gen data center demands. Up to date, many modulators of different structures have achieved high electro-optical bandwidth and transmission rate. High electro-optical bandwidth optical modulator above 110 GHz and high speed optical modulator above 300 Gb/s has been demonstrated using Mach-Zehnder modulators (MZMs) [10,11]. However, MZMs typically require several millimetres in length, which severely limits the design and expansion of high-density optical chips, and also causes high power consumption to be difficult to be compatible with CMOS circuit drivers. Whereas micro-ring modulators (MRMs) have smaller footprint, lower power consumption and a modulation efficiency comparable to MZMs [12]. Therefore, MRM is considered to realize the photoelectric signal conversion function in the high density photoelectric interconnection system. The electro-optic (EO) modulation of the Si MRMs can be realized mainly by adjusting the carrier concentration

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with the free carrier dispersion effect. According to the doping structure of the modulator and the different electrical working conditions, the modulator can be divided into carrier injection, accumulation, and depletion modulator [13].

The carrier injection mode Si MRM is composed of a p-i-n structure, and the refractive index is changed by forward injection of carriers. Although it can achieve high modulation efficiency, its operating speed is limited by the lifetime of silicon carriers, making it difficult to achieve high rate transmission [14]. The carrier accumulation mode MRM uses a forward-biased metal-oxide semiconductor (MOS) capacitor with a thin insulating oxide layer sandwiched between the P-type polysilicon and N-type silicon to form a waveguide, so that the light field and the cumulative carrier variation region overlap. This structure enables efficient modulation, and the bandwidth is not limited by the carrier lifetime, but is controlled by the RC constant of the MOS capacitor [15]. Unlike the first two, the carrier depletion mode MRM uses the reverse bias voltage to change the waveguide refractive index. The reverse-biased p-n junction relives the operating speed constraint of the minority carrier lifetime, resulting in faster bandwidth and signal transfer rates. The first silicon-based MRM was proposed in 2005, incorporating a lateral PIN structure [14]. Subsequently, various MRMs with different structures have been reported. Due to the mechanism limitation, most high speed MRMs are depletion mode PN junction modulators [16–18]. However, the depletion MRM has lower modulation efficiency than the other two Si MRM modes, led to the practical operation typically necessitates a drive voltage (Vpp) exceeding 2 V. Various PN junction structures have been developed for the depletion mode modulator to enhance the overlap factor between the carrier variation region and light field to improve modulation efficiency, including the vertical junction [19–22], lateral junction [23–29], L-shape junction [30–32] and U-shape junction [33]. But in fact, the performance of silicon depletion mode MRM is still limited by the trade-off between modulation efficiency and RC time-limited bandwidth.

Here we report a depletion mode Si MRM with a meandering doping PN junction. The meandering doping can realize lateral PN junction and longitudinal PN junction to improve modulation efficiency while the junction capacitance is not significantly affected. This MRM exhibits a high modulation efficiency of 33.5 pm/V, a high EO bandwidth above 67 GHz and an energy consumption of 8.54 fJ/bit for 128 Gb/s non-return-to-zero (NRZ) modulation. The experiments show that the modulator has a good prospect for application in optical interconnection system.

2. Modulator design and fabrication

The silicon micro-ring modulator was fabricated on a commercial foundry service multi-project wafer (MPW) silicon-on-insulator (SOI) platform. The modulator has two strip ridge waveguides as an add-drop modulator, light is coupled from the grating coupler of the input port into the modulator, the through port is used to transmit optical signals and the drop port is connected to a high-speed photodetector (PD) for wavelength control or signal monitoring. The structure of device is shown in Fig. 1(a). The MRM was formed by a silicon rib waveguide with height of 220 nm, width of 500 nm, slab thickness of 90 nm and buried oxide of 2 µm. The radius of the MRM is designed to be 8 µm. The gap between through port waveguide and ring-ridged waveguide is gap1 and the gap between drop port waveguide and ring-ridged waveguide is gap2. The extinction ratio and quality factor of the modulator can be controlled by adjusting the two gaps separately. In our design, gap1 and gap2 is 200 nm and 260 nm respectively. Closer gap1 is needed to achieve high coupling efficiency between through port waveguide and ring-ridged waveguide, while further gap2 is to allow the ring to work closer to the critical coupling state. A layer of TiN heater is grown to use the silicon's thermo-optical effect to tune the resonance wavelength of the MRM. The metallographic microscope image of the MRM is shown in Fig. 2.

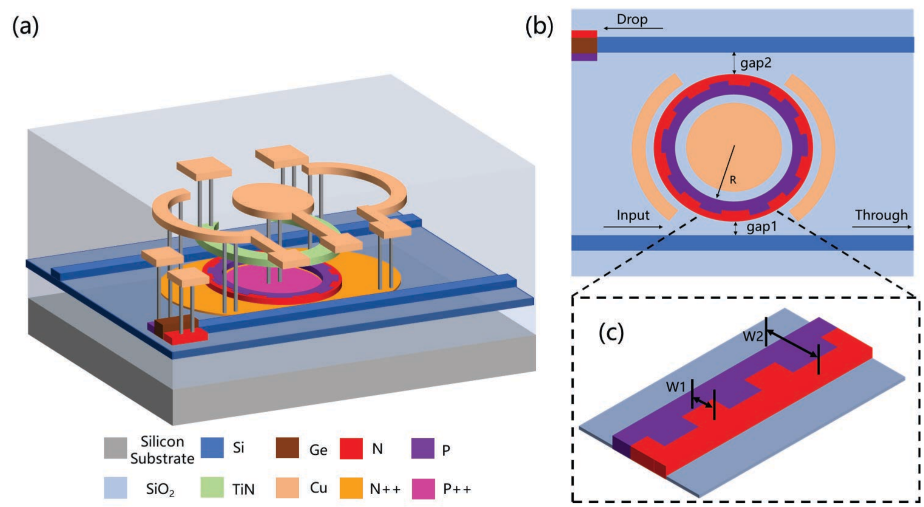

Fig. 1. (a) Schematic overview of the designed MRM structure, (b) Overhead structure chart of the MRM, (c) Schematic overview of the meandering doping structure.

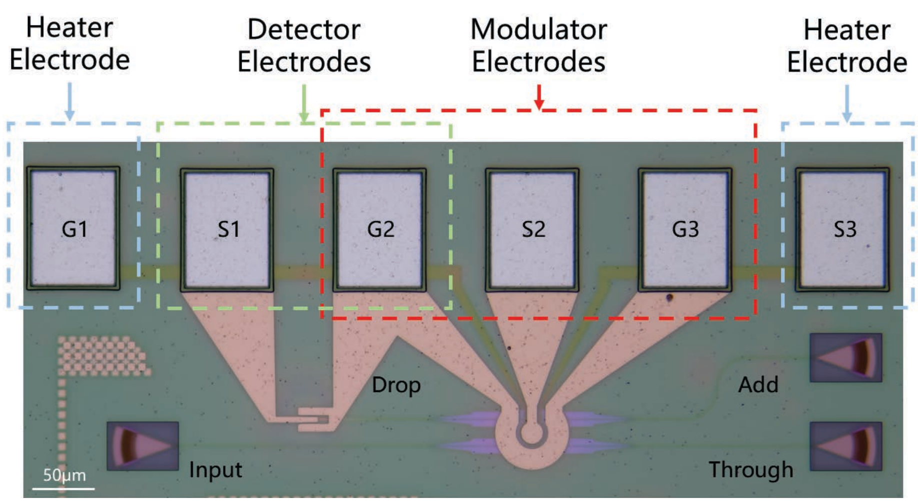

Fig. 2. Microscope image of the fabricated MRM.

The bond pads are designed for dc signal and ac signal loading, each pad has dimension of $60 \times 80\ \mu m^2$ and the pad spacing is 100 μm.

The traditional MRMs often use lateral PN junction with depletion mode and their modulation efficiency is basically lower than 30 pm/V [18], so we design a meandering doping structure to improve the modulation efficiency of the MRMs which is suitable for commercial active process silicon based device chip factory. The meandering doping design depicted in Fig. 1(b) and Fig. 1(c) showcase a winding pathway, where the P-type doping region is located inside the ring-ridged waveguide, the N-type doping region is located outside the ring-ridged waveguide, and the middle region of the ring-ridged waveguide adopts P-doping and N-doping meandering doping. The whole junction has four doping levels for P doping and N doping. The lower concentration was used at ring-ridge waveguide region to reduce propagation loss and the higher concentration was used at slab region for metallic contact. The meandering doping structure is

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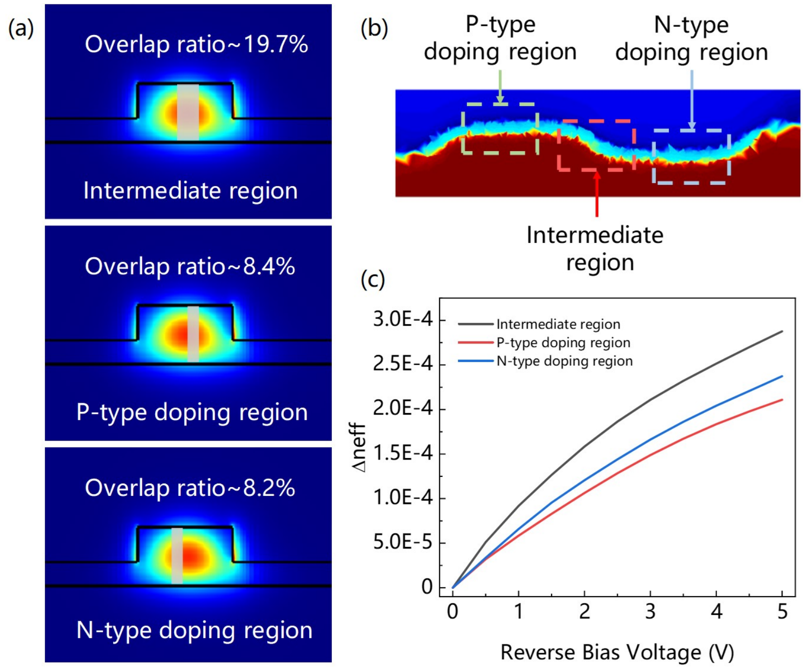

Fig. 3. (a) Calculated overlap ratios between the micro-ring waveguide transverse electric (TE) mode and depletion regions of meandering doping different regions, (b) Simulation results of carrier doping distribution at meandering doping regions, (c) Effective refractive index real part versus reverse bias variation curve of meandering doping different regions.

arranged alternately with different block-type doping, W1 is the transverse width of meandering doping W1 with 100 nm while W2 is the transverse width of micro-ring waveguide with 500 nm. We divided the meandering doping of the middle 100 nm width into three main parts, the intermediate depletion region, the p-type doping region and the n-type doping region. The intermediate depletion region is longitudinal PN junction due to the different doping types before and after, the width of the depletion region is the cross-sectional width W1. Using the mode module of the simulation software, calculated overlap ratios between the micro-ring waveguide transverse electric (TE) mode and depletion regions of the intermediate depletion region is 19.7% as shown in Fig. 3(a). While the depletion region of the other two parts, the p-type doping region and the n-type doping region are formed by different types of doping on the left and right sides, the overlap factors calculated by assuming the width of the depletion region to be 40 nm are 8.4% and 8.2%, respectively. The above calculation yields a larger overlap factor in the intermediate depletion region and therefore a higher modulation efficiency as well. We simulated the variation of carrier distribution in the micro-ring waveguide with voltage change. In Fig. 3(b) we can see the meandering doping regions can realized lateral PN junction and longitudinal PN junction, which can change simultaneously with the change of PN bias voltage. Since the overlap factor between the depletion region and the optical field region of the longitudinal PN junction is larger than lateral PN junction, and the proportion of the longitudinal PN junction to the overall PN junction is small without affecting the PN junction capacitance, finally the overlap factor between the depletion region and the optical field region of the ring ridge waveguide can be increased, so as to improve the modulation efficiency of the MRM without reducing the device bandwidth. Hybrid calculations of carrier variations with bias voltage in the doped region and waveguide optical field patterns to obtain waveguide refractive index variations at different bias voltages.

The intermediate depletion region has a higher refractive index change efficiency resulting in higher modulation efficiency as illustrated in Fig. 3(c).

3. Transmission characterization

Transmission performance of the meandering doping micro-ring modulator (MD-MRM) at no bias voltage is depicted in Fig. 4(a). The extinction ratio (ER) of our design was measured to be 9 dB, which means the MRM was working in the under coupling state. Consequently, we can reduce gap1 and change gap2 to make the MRM work closer to the critical coupling state next time. The measured free spectral range (FSR) is 11.75 nm, due to the problem of grating coupler fabrication, the FP cavity effect of the grating causes serious oscillation of the transmission spectrum lines of the MRM in the region of wavelength greater than 1555 nm. We used the optical sweep system and DC power supply to coordinate the test, through the RF probe to export the drop port detector photocurrent. The current response diagram of the detector monitored varied with the wavelength of light is depicted in Fig. 4(a), the wavelength corresponding to the peak of current change is basically the same as the resonant wavelength of the MRM. The result indicates that the drop port detector can realize the function of wavelength detection. Because of the process preparation tolerance, the MRM is difficult to work directly at 1550 nm or other wavelength we design, the resonance wavelength of the MRM can be tuned by the layer of TiN heater with thermo-optical effect. The applied voltage at heater pads and the MRM wavelength tuning is shown in Fig. 4(b), and the tuning efficiency is 75 pm/mW. We set up a control group with the same device structure parameters except for the doping structure, which used lateral doping. The lateral doping micro-ring modulator (LD-MRM) and MD-MRM transmission performance of the through port were measured by the tunable laser sweep frequency system. Figure 5(a) and (b) show the measured transmission spectrum of two MRMs with PN biasing voltages from 0 V to – 4 V. The EO modulation efficiency of the MD-MRM was measured to be 33.5 pm/V while the LD-MRM was 24.3 pm/V in Fig. 5(c). Compared with LD-MRM, the MD-MRM's EO modulation efficiency can improve by about 38%. It is proved that meandering doping can indeed improve the modulation efficiency. The loaded Q factor of the two MRMs were measured to be ~5000 and ~5200. The Fig. 5 illustrates that, meandering doping has little effect on the capacitance of the modulator.

Figure 6(a) shows fitted equivalent circuits and corresponding junction resistance-capacitance (RC) responses for MD-MRM at -3 V. The results were fitted to a simple linear equivalent circuit model for depletion type micro-ring modulator [28]. C_{pad} is capacitance formed between signal

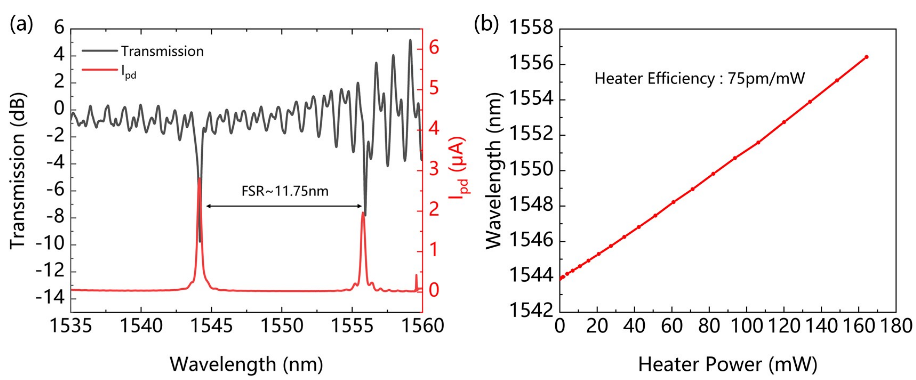

Fig. 4. (a) Transmission performance of the MD-MRM at no bias voltage and the current response diagram of the detector monitored by the drop port, (b) The resonance wavelength of the MD-MRM versus heater power.

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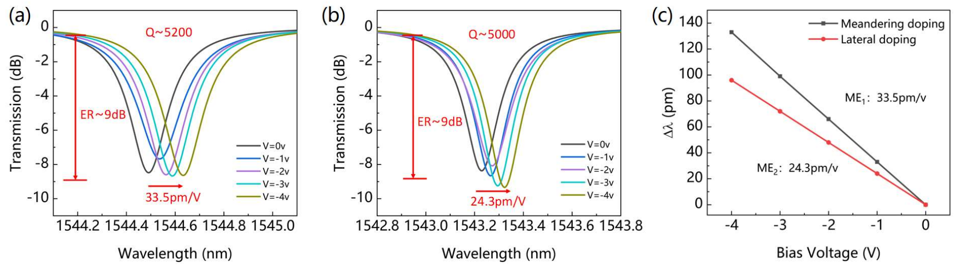

Fig. 5. Transmission performance of (a) MD-MRM, (b) LD-MRM with PN biasing voltages from 0 V to – 4 V, (c) Extracted resonance wavelengths versus junction voltage and calculated modulation efficiency (ME) for MD-MRM and LD-MRM.

electrodes, R_{sub} is the silicon substrate leakage and C_{sub} is capacitance formed from electrode to silicon substrate. R_{pn} and C_{pn} are the serial resistance and junction capacitance of the MRM. The calculation obtained by fitting the S_{11} curve with the above five parameters of the equivalent circuit, shown the MRM had a serial resistance of 59.47 ohms and a junction capacitance of 18.48 fF. According to the capacitor resistance value of the modulator, the electrical bandwidth of the modulator is greater than 100 GHz. It means that the EO bandwidth of the MRM is mainly limited by the optical bandwidth rather than the electrical bandwidth. The EO response (S_{21}) was measured using a lightwave component analyzer (LCA; Keysight N4372D) with a 110 GHz bandwidth. The instrument was calibrated using a ceramic calibration pieces to eliminate the effects of high-speed probes, RF adapters, and cables. The laser output light through the fiber passed through the polarization controller and was coupled into the modulator waveguide using the grating coupler on the chip. The DC bias voltage and the small signal output from the LCA were loaded on the high-speed probe through the bias tee at the same time and tied to the GSG electrode of the MRM. The output of the modulator was amplified by the erbium-doped fiber amplifier (EDFA) and fed into the LCA. The laser wavelength was 1544.1 nm off the resonant wavelength of MRM, when the PN bias was -1 V to -6 V, the MRM shown a high 3 dB EO bandwidth more than 67 GHz, as shown in Fig. 6(b). The optical peaking effect can significantly affect the EO bandwidth of the modulator, and the effect changes with the working wavelength of the modulator [34]. The EO responses were measured to reach 110 GHz with -5 V bias voltage. Figure 6(c) shows the bandwidth measurements made at different operation wavelength under -3 V bias voltage. The result show that when the TL wavelength drop from 1544.4 nm, the 3 dB EO bandwidth of the MRM gradually increased and the bandwidth downward trend of 1544.1 nm is slower than that of 1544.4 nm. The optical peaking effect is strong when the MRM operation wavelength is far from resonance wavelength.

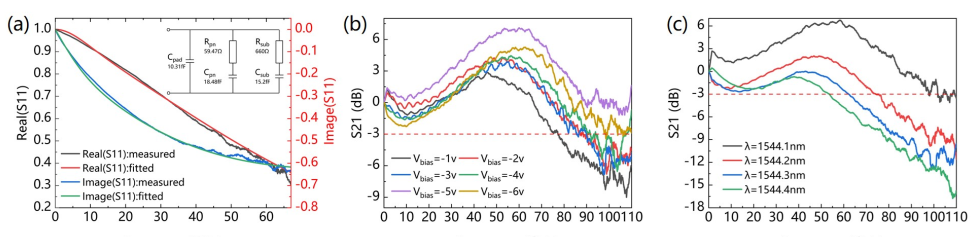

Fig. 6. (a) Fitted equivalent circuits and corresponding junction resistance-capacitance (RC) responses for MD-MRM at -3 V. (b) Measured EO response (S21) at different bias voltage, (c) Measured EO response (S21) at different working wavelength under -3 V bias voltage.

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4. High-speed characterization

The high-speed communications performance was tested using the large signal test system as shown in the Fig. 7. The output light from the tunable laser was transmitted to the polarization controller (PC) and coupled into the input port of MRM through the grating coupler on the chip. The PC is used to adjust the polarization state of the input light to the TE mode which can reduce polarization-dependent loss. The output power of the tunable laser was 10 dBm and the operation wavelength was directly adjusted by the laser instead of changing resonance wavelength of the MRM. We used an arbitrary waveform generator (AWG; Keysight M8199A) to generate the RF signal. The AWG has a nominal sampling rate of 128 GSa/s which can support up to 128 Gbaud NRZ modulation signal transmission. The high-speed RF signal output of AWG was amplified using RF amplifier (N4985A) and the peak-to-peak voltage of the RF signal transmitted to the MRM was lower than 2 V. We used a bias tee to load both DC bias and RF electrical signal and utilized a 67 GHz GSG probe tied to the GSG electrode of the MRM. The output of the modulator was amplified by the erbium-doped fiber amplifier (EDFA) to compensate for fiber–chip coupling loss, then a tunable optical bandpass filter (OBPF) is utilized to attenuate the amplified spontaneous emission (ASE) noise. In the end, the light signal was fed into a digital communications analyzer (DCA; Keyisght N1000A). The DCA has a built-in high-speed photodetector to convert the optical signal into the electrical signal and was used for digitizing the electrical signal from photodetector.

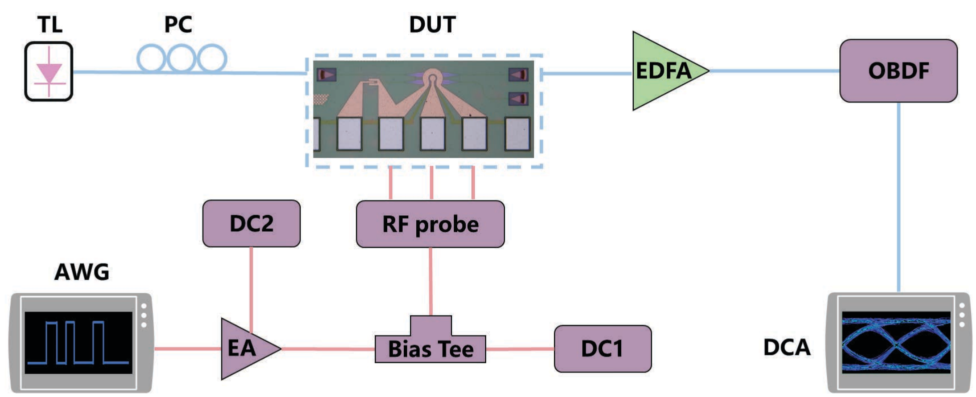

Fig. 7. Experimental setup of the transmission based on the MD-MRM. TL, tunable laser; PC, polarization controller; DUT, device under test; AWG, arbitrary wave generator; EA, electrical amplifier; DC, dc regulated power; EDFA, erbium-doped fiber amplifier; OBPF, optical bandpass filter; DCA, digital communications analyzer.

In the high-speed communications performance test, the bias voltage of the PN junction was -3 V. We tested the eye diagram and extinction ratio obtained by DCA at different rates from 25 Gb/s to 128 Gb/s. At the signal receiving port, feed-forward equalizer (FFE) algorithm was used to equalize the actual collected signal waveform to improve the signal quality. Figure 8 shows different open eye diagram obtained by DCA at different NRZ modulation signal transmission rates under -3 V PN junction bias voltage. The measured eye diagrams exhibit clear eye openings, which confirms that the MRM can be employed as a high-speed electro-optic modulator. While the transmission bit rate increases, the eye diagram open less with ER decreases and the stitch widens. We obtained an open eye with ER of 4.80 dB, 4.70 dB, 3.26 dB, 3.45 dB, 3.53 dB, 2.90 dB at 25 Gb/s, 56 Gb/s, 76 Gb/s, 96 Gb/s, 110 Gb/s 128 Gb/s, respectively. The drive voltage of the MRM loading 128 Gb/s signal is only 1.36 V, so that the power consumption at 128 Gb/s is calculated to be ~8.54 fJ/bit. For high-speed signal transmission, in addition to the device electro-optical bandwidth limitation, RF amplifier output voltage swing Vpp is also gradually reduced when the signal transmission rate increases. All these will lead to the deterioration of quality of the eye diagrams at the receiving end. The 256 GS/s real-time digital storage oscilloscope (DSO) was used to capture the electrical signal output by the photodetector, and the bit error rate (BER) was calculated by offline digital signal processing (DSP). Figure 9 shows the measured BER for back to back (BtB) transmission of the MRM with different received optical power for 100, 112, 120, 128 Gb/s NRZ signal. The calculated BER of 128 Gb/s NRZ at 4dBm received optical power is 0.9×10^{-2}. At the same transmission signal rate, the BER will increase with the decrease of the received optical power, and the BER will also increase with the increase of the transmitted signal rate at the same received optical power. When the 100 Gb/s transmission rate at -4 dBm receiving optical power, 112 Gb/s rate at -2 dBm, 120 Gb/s rate at -1 dBm, and 128 Gb/s rate at 1dBm, the calculated BER of NRZ signal is still lower than the 2×10^{-2} soft-decision forward error correction (SD-FEC) assumption threshold. In addition, we conducted a high-speed test of pulse amplitude modulation with four levels (PAM-4). The signal eye diagrams of PAM-4 signal of 56 Gb/s, 100 Gb/s, 160 Gb/s, 200 Gb/s are shown in Fig. 10.

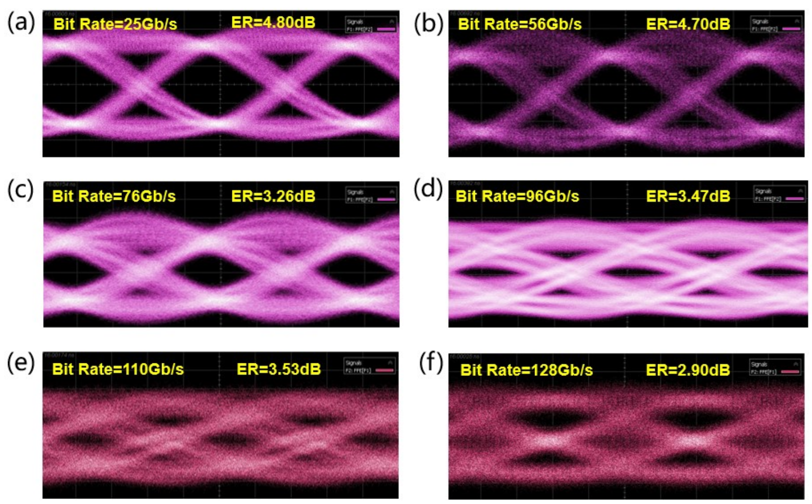

Fig. 8. Measured NRZ modulation eye diagrams at (a) 25 Gb/s, (b) 56 Gb/s, (c) 76 Gb/s, (d) 96 Gb/s, (e) 110 Gb/s, (f) 128 Gb/s.

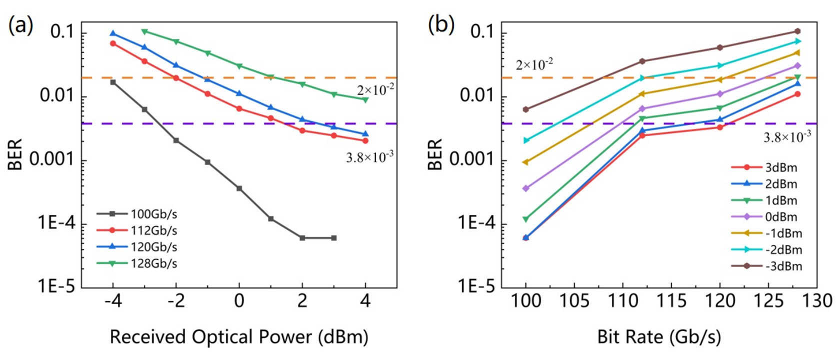

Fig. 9. Measured BER for BtB transmission of the MRM (a) with different received optical power for 100, 112, 120, 128 Gb/s NRZ signal, (b) with different transmission rate NRZ signal.

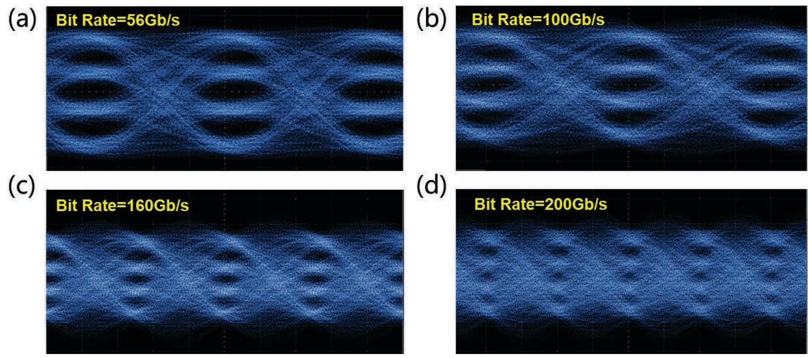

Fig. 10. Measured PAM-4 modulation eye diagrams at (a) 56 Gb/s, (b) 100 Gb/s, (c) 160 Gb/s, (d) 200 Gb/s.

5. Discussion

Table 1. Properties of different type of Si depletion mode PN junction MRMs

Ref	Doping Structure	Radius (μm)	BW (GHz)	Modulation Efficiency (pm/V)	Operating Wavelength (nm)	Bit Rate (Gb/s)	Vpp (V)	Energy Cost (fJ/bit)
16	Z-shape	12	58	27.3	1310	200 (PAM-4)	1.6	6.3
18	Lateral	8	>60	26.4	1310	240 (PAM-8)	3	-
22	vertical	5	24	58.1	1310	40 (NRZ)	1	4
25	Zigzag-shape	10	>40	16	1550	44 (NRZ)	3	300
32	L-shape	4	54	38	1310	240 (PAM-4)	1.8	-
33	U-shape	65	13.5	40	1310	13 (NRZ)	1.6	-
This work	Meander-ing-shape	8	>67	33.5	1550	128 (NRZ)	1.36	8.54

Due to the problem of grating coupler fabrication, the center wavelength of the grating coupler is near 1520 nm, far away from 1550 nm, resulting in a large loss of the grating coupler itself in the C-band, the transmission loss near resonance wavelength 1544.4 nm of using two vertical gratings to input and output is approximately 13.5 dB. And the FP cavity effect of the grating causes serious oscillation of the transmission spectrum lines of the MRM in the region of wavelength greater than 1555 nm. There are a number of solutions available for high loss vertical gratings, such as apodized bidirectional grating [35], genetic algorithm optimized grating [36], resonance-enhanced wideband grating [37] and so on. Excessive loss will lead to low light signal output power of the MRM, which increases the difficulty of oscilloscope signal recognition and increases noise and bit error rate. In addition, the bandwidth of other instruments in the test system and the processing algorithm of the received signal will also affect the test results. Therefore, we can improve the NRZ or PAM-4 signal transmission quality by reducing the overall optical loss of the test system, using higher bandwidth instruments and more accurate and reproducible algorithm.

A detailed analysis has been included, showcasing a thorough comparison between different MRMs in Table 1. As observed, the proposed MRM exhibit a higher electro-optic modulation efficiency and a low drive voltage while achieving high transmission rates. Meanwhile the MRM can be readily manufactured using the conventional MPW foundry procedure, which is beneficial to ensure large-scale preparation and application.

Furthermore. since the Vpp of the MRM loading 128 Gb/s signal is only 1.36 V, it can be considered to apply the MRM in the optical interconnection chip, and use the electrical chip to directly drive the MRM, and finally realize the high-density optical interconnection compatible with the CMOS chip. The modulator, as the core device for photoelectric information conversion, necessitates a high bandwidth and low drive voltage that aligns with the CMOS drive voltage. We posit that the performance of the MRM holds potential for application in this domain.

6. Conclusion

We obtained a high electro-optic modulation efficiency MRM with high bandwidth and low drive voltage. The measured modulation efficiency of the MRM is 33.5 pm/V, compared with the ordinary lateral MRM 24.3 pm/V, the modulator efficiency is improved by above 38%. Under -3 V PN junction bias voltage, the MRM shows above 90 GHz bandwidth. In back-to-back high-speed transmission test, 128 Gb/s NRZ transmission was achieved with 2.90 dB ER, and the drive voltage of the MRM was 1.36 V, the corresponding power consumption is ~8.54 fJ/bit. The small drive voltage and low energy cost enable the MRM to be electrically driven compatible with CMOS circuits in addition to being compatible with CMOS platform materials and processes. The proposed silicon MRM can play a crucial role in the high density and high-speed photoelectric interconnection with its high bandwidth and low drive voltage.

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Disclosures. The authors declare no conflicts of interest.

Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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